

## EMCH 201: Introduction to Application of Numerical Methods

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### Assignment 4

Assignment due date: **11:59 PM, 1<sup>st</sup> Nov, 2021**

**Point Possible: 100 (20+40+40)**

#### Submission requirements:

- Detailed problem-solving process needs to be provided for full points
- Late submission will not be graded

1. Given these data points

8.9    9.5    9.8    9.5    10.1    9.3

Determine: (a) the mean, (b) the standard deviation, (c) the variance, and (d) the coefficient of variance. (Note: Show essential steps of calculation). (20 points)

2. An investigator has reported the data tabulated below for an experiment to determine the growth rate of bacteria  $k$  (per d), as a function of oxygen concentration  $c$  (mg/L), i.e.,  $k = f(c)$

|   |      |      |      |      |      |
|---|------|------|------|------|------|
| c | 0.5  | 0.8  | 1.6  | 2.5  | 4    |
| k | 1.08 | 2.41 | 5.35 | 7.91 | 8.95 |

It is also known that such data can be modeled by the following equation:

$$k = \frac{k_{\max} c^2}{c_s + c^2}$$

Compute coefficients  $k_{\max}$  and  $c_s$ , the sum of the squares of the regression residual  $S_r$ , and the coefficient of determination  $r^2$ . **Hint:** Use the reciprocal form we learned in the class to transform and linearize the growth rate model above. (Note: Show all the steps of calculation and you can write your own program in Matlab to compute the various summations of data). (40 points)

3. A function follows the tabulated data below:

|   |       |       |       |       |    |
|---|-------|-------|-------|-------|----|
| x | -1.80 | -1.65 | -1.50 | -1.25 | -1 |
| y | -40   | -20   | 10    | 38    | 76 |

Fit the data to a 2<sup>nd</sup>-order polynomial, i.e.,  $y = a_0 + a_1x + a_2x^2$  and do the following. (Note: Show all the steps of manual calculation).

- Calculate all the necessary quantities to be used for the 2<sup>nd</sup>-order polynomial regression, including  $\sum x_i$ ,  $\sum x_i^2$ ,  $\sum x_i^3$ ,  $\sum x_i^4$ ,  $\sum x_i y_i$ ,  $\sum x_i^2 y_i$ ,  $\sum y_i$  (10 points)
- Put the regression equations into the matrix form. (5 points)
- Solve the linear equations and Present your regression coefficients clearly. (15 points)
- Calculate the sum of the squares of the regression residual  $S_r$ , and the coefficient of determination  $r^2$ . (10 points)